

MAT 524: Functions of a Real Variable II

Homework VI

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[[Problem One]]:

Problem Statement

Let H be a Hilbert space. For $y \in H$, define $T_y \in H^*$ by

$$T_y(x) = \langle x, y \rangle.$$

Prove that $\|T_y\| = \|y\|$. Explain why it follows that the map $\phi : H \rightarrow H^*$ defined by $y \mapsto T_y$ is one-to-one.

Solution

Let $y \in H$ and let $S = \{M : |T_y(x)| \leq M \|x\| \text{ for all } x \in H\}$. Note that $\|y\| \in S$, by the Cauchy-Schwarz inequality. Thus, to show that $\|y\| = \inf S = \|T_y\|$, we simply need to show that $\|y\|$ is a lower bound for S . Suppose otherwise. Then there exists $M \in S$ such that $M < \|y\|$ and $|T_y(x)| = |\langle x, y \rangle| \leq M \|x\|$ for all $x \in H$. But $|\langle y, y \rangle| = \|y\|^2 > M \|y\|$. This contradicts the fact that $M \in S$. Thus, $\|y\| = \inf S = \|T_y\|$.

To show that ϕ is one-to-one, let $y, z \in H$ and suppose $\phi(y) = \phi(z)$. Then for all $x \in H$, we have $\langle x, y \rangle = \langle x, z \rangle$. In particular, $\langle y - z, y \rangle = \langle y - z, z \rangle$, or

$$0 = \langle y - z, y \rangle - \langle y - z, z \rangle \tag{1}$$

$$= \langle y - z, y \rangle + \overline{(-1)} \langle y - z, z \rangle \tag{2}$$

$$= \langle y - z, y \rangle + \langle y - z, -z \rangle \tag{3}$$

$$= \langle y - z, y - z \rangle. \tag{4}$$

But this implies that $y - z = 0$, i.e., $y = z$. Therefore, ϕ is one-to-one. ■



[[Problem Two]]:

Problem Statement

Let M be a closed subspace of a Hilbert space H . Prove that $M = (M^\perp)^\perp$.

Solution

Let $x \in M$. Then for all $y \in M^\perp$, we have $\langle x, y \rangle = 0$. But if $\langle x, y \rangle = 0$ for all $y \in M^\perp$, then $x \in (M^\perp)^\perp$.

Thus $M \subseteq (M^\perp)^\perp$. Next, let $x \in (M^\perp)^\perp$. Since M is a closed subspace of H , we can uniquely decompose x as $x = a + b$, where $a \in M$ and $b \in M^\perp$. Then $\langle x, b \rangle = 0$ and $\langle a, b \rangle = 0$, so

$$\langle x, b \rangle - \langle a, b \rangle = \langle x - a, b \rangle \tag{5}$$

$$= \langle b, b \rangle \tag{6}$$

$$= 0. \tag{7}$$

But then $b = \vec{0}$, i.e., $x = a$. Therefore, $x \in M$, which means $(M^\perp)^\perp \subseteq M$. Thus, $M = (M^\perp)^\perp$.

■



[[Problem Three]]:

Problem Statement

If M is an arbitrary subset of H , prove that $(M^\perp)^\perp$ is the smallest closed subspace containing M .

Solution

(i) We know from the previous problem that $M \subseteq (M^\perp)^\perp$.

(ii) To show that $(M^\perp)^\perp$ is a subspace, we first note that $\vec{0} \in (M^\perp)^\perp$ (since the inner product of anything with $\vec{0}$ is 0), so $(M^\perp)^\perp$ is nonempty. Let $x, y \in (M^\perp)^\perp$. Then for all $z \in M^\perp$, we have $\langle x, z \rangle = 0$ and $\langle y, z \rangle = 0$. But then

$$\langle x, z \rangle + \langle y, z \rangle = \langle x + y, z \rangle \quad (8)$$

$$= 0. \quad (9)$$

Thus, $x + y \in (M^\perp)^\perp$. Next, let $\lambda \in \mathbb{C}$. Since $\langle x, z \rangle = 0$, it follows that

$$\langle \lambda x, z \rangle = \lambda \langle x, z \rangle \quad (10)$$

$$= \lambda \cdot 0 \quad (11)$$

$$= 0. \quad (12)$$

Hence, $\lambda x \in (M^\perp)^\perp$. Therefore, $(M^\perp)^\perp$ is a subspace.

(iii) Next, we show that $(M^\perp)^\perp$ is closed. Let $\{x_n\}$ be a sequence in $(M^\perp)^\perp$ that converges to x . We will show that $x \in (M^\perp)^\perp$. Let $y \in M^\perp$. We have

$$\langle x, y \rangle = \langle \lim x_n, y \rangle \quad (13)$$

$$= \lim \langle x_n, y \rangle \quad (14)$$

$$= \lim 0 \quad (15)$$

$$= 0. \quad (16)$$

Therefore, $x \in (M^\perp)^\perp$, meaning $(M^\perp)^\perp$ is closed.

Finally, we will show that $(M^\perp)^\perp$ is the smallest set satisfying (i) - (iii). Suppose S is a set satisfying (i)-(iii) that contains M . If $x \in S^\perp$, then x is orthogonal to everything in S . But then x must be orthogonal to everything in M , as well, so $x \in M^\perp$, i.e., $S^\perp \subseteq M^\perp$. By similar reasoning, $(M^\perp)^\perp \subseteq (S^\perp)^\perp$. Now, let $y \in (S^\perp)^\perp$. Since S is a closed subspace, we can write $y = a + b$, where $a \in S$ and $b \in S^\perp$. It must be the case that $y \perp b$ and $a \perp b$. So

$$\langle y, b \rangle = \langle a + b, b \rangle \tag{17}$$

$$= \langle a, b \rangle + \langle b, b \rangle \tag{18}$$

$$= \langle b, b \rangle \tag{19}$$

$$= 0. \tag{20}$$

But then $b = \vec{0}$. So $y = a$, which means $y \in S$. Thus, $(M^\perp)^\perp \subseteq (S^\perp)^\perp \subseteq S$. Therefore, $(M^\perp)^\perp$ is the smallest set satisfying (i) - (iii).

■



[[Problem Four]]:

Problem Statement

Let M be a closed subspace of a Hilbert space H , and let $P : H \longrightarrow M$ denote the orthogonal projection onto M .

(a) Prove that (i) P is continuous, (ii) $P^2 = P$, and that (iii) P is self-adjoint, i.e. $\langle Px, y \rangle = \langle x, Py \rangle$ for all $x, y \in H$.

(b) Conversely, if $Q : H \longrightarrow H$ is a continuous linear operator satisfying $Q^2 = Q$ and which is self-adjoint, prove that the image $Q(H)$ of Q is a closed subspace, and that Q is the orthogonal projection onto $Q(H)$.

Solution

(a) (i) Let $x, y \in H$. Since M is a closed subspace of H , we can write $x = a + b$, and $y = c + d$, where $a, c \in M$ and $b, d \in M^\perp$. Then $x - y = (a - c) + (b - d)$, and since M and $M^\perp$ are closed subspaces, we have $(a - c) \perp (b - d)$. Thus, we can apply the Pythagorean Theorem to obtain

$$\|x - y\|^2 = \|a - c\|^2 + \|b - d\|^2. \quad (21)$$

But then

$$\|a - c\| = \sqrt{\|x - y\|^2 - \|b - d\|^2} \quad (22)$$

$$\leq \sqrt{\|x - y\|^2} \quad (23)$$

$$= \|x - y\|. \quad (24)$$

Now, let $\varepsilon > 0$. Choose $\delta = \varepsilon$. Then whenever $\|x - y\| < \delta$, we have

$$\|Px - Py\| = \|a - c\| \quad (25)$$

$$\leq \|x - y\| \quad (26)$$

$$< \delta \quad (27)$$

$$= \varepsilon. \quad (28)$$

Therefore, P is continuous.

(ii) Let $x \in H$, where $x = y + z$, with $y \in M$ and $z \in M^\perp$. Then

$$P^2x = P(Px) \quad (29)$$

$$= Py. \quad (30)$$

But since $y \in M$, we have $Py = y = Px$. Thus, $P^2 = P$.

(iii) Let $x, y \in H$, where $x = a + b$ and $y = c + d$, with $a, c \in M$ and $b, d \in M^\perp$. We have

$$\langle Px, y \rangle - \langle x, Py \rangle = \langle a, c + d \rangle - \langle a + b, c \rangle \quad (31)$$

$$= \langle a, c \rangle + \langle a, d \rangle - \langle a, c \rangle - \langle b, c \rangle \quad (32)$$

$$= \langle a, d \rangle - \langle b, c \rangle \quad (33)$$

$$= 0 - 0 \quad (\text{since } a, c \in M \text{ and } b, d \in M^\perp) \quad (34)$$

$$= 0. \quad (35)$$

Thus, $\langle Px, y \rangle = \langle x, Py \rangle$.

(b) We begin by showing that $Q(H)$ is a subspace of H . Since $\vec{0} \in H$, $Q(\vec{0}) \in Q(H)$. Thus, $Q(H)$ is nonempty. Let $x, y \in Q(H)$. Then there exist $a, b \in H$ such that $Q(a) = x$ and $Q(b) = y$. Since H is closed, $a + b \in H$, so $Q(a + b) \in Q(H)$. But $Q(a + b) = Qa + Qb = x + y$. Therefore, $x + y \in Q(H)$. Let $\lambda \in \mathbb{C}$. Then $\lambda a \in H$. So $Q(\lambda a) = \lambda Q(a) = \lambda x \in Q(H)$. Hence, $Q(H)$ is a subspace of H .

To show that $Q(H)$ is closed, let $\{x_n\}$ be a sequence in $Q(H)$ that converges to x . For each $n \in \mathbb{N}$, there exists $a_n \in H$ such that $Q(a_n) = x_n$.

Since $Q(H)$ is a closed subspace of H , for any $x \in H$, we can write $x = y + z$, where $y \in Q(H)$ and $z \in Q(H)^\perp$. We want to show that $Qx = y$. This will be achieved if we can show that $\langle Qx - y, Qx - y \rangle = 0$.

■

